# OpenClaw Plant Care Assistant Tutorial

> Before starting, please complete [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [05-AI Voice Interaction Configuration], and [06-Three Dialogue Configuration Tests] in the `Pre-use Preparation` directory. If you are only using TUI / WhatsApp, you can skip `05` for now. This article assumes the basic environment is already prepared and only covers the plant care project itself.

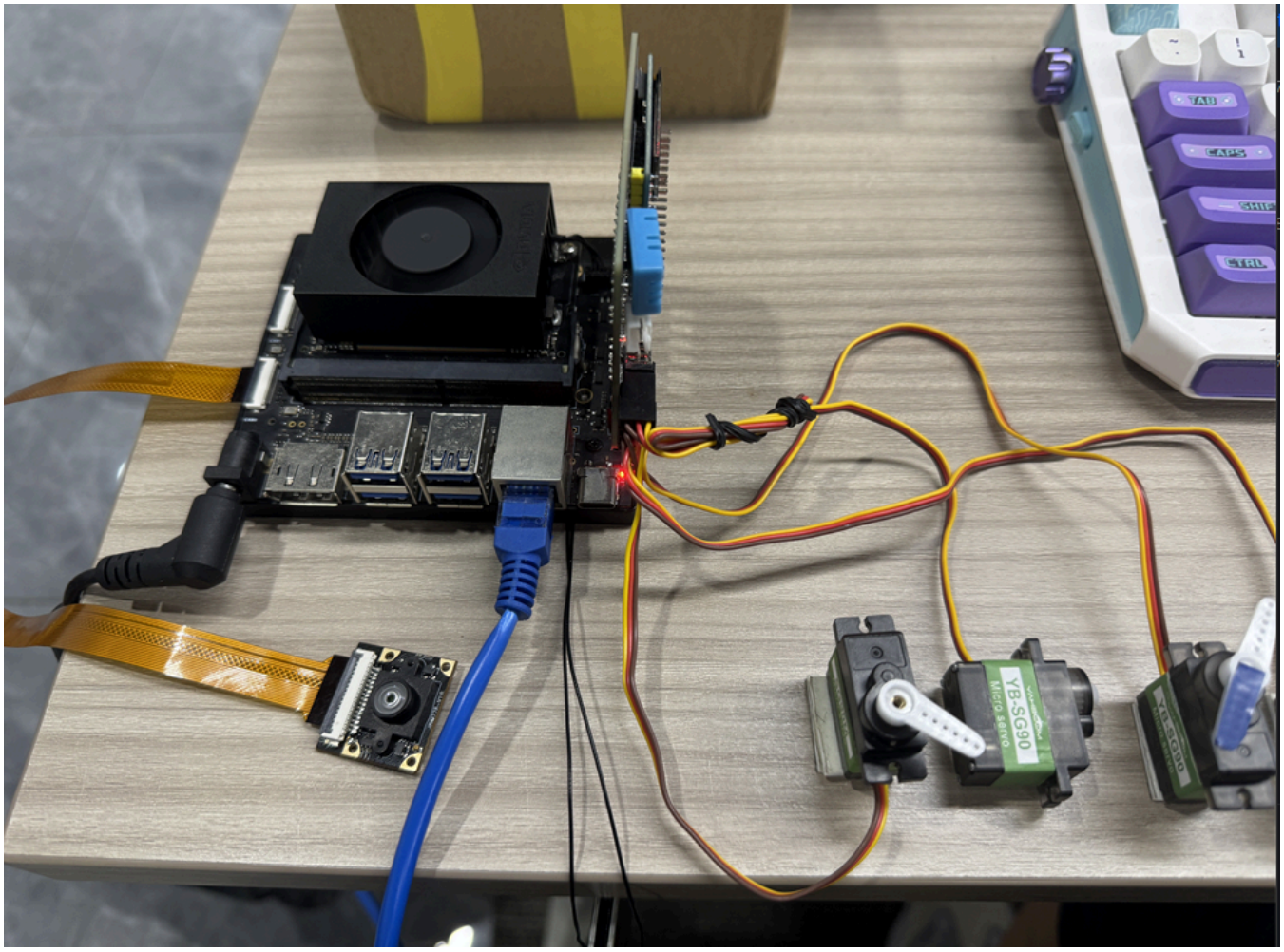
## 1. Tutorial Objective

This article demonstrates OpenClaw's multi-peripheral collaboration capability through a plant care scenario.

- Learn how to combine temperature/humidity sensors, OLED, RGB, and camera into a plant care assistant
- Interact with OpenClaw through three methods: WhatsApp, TUI, and voice interaction

## 2. Hardware Preparation

- OpenClaw main control board
- DHT temperature/humidity sensor
- OLED display
- RGB light strip / RGB LED module
- CSI camera
- `[Optional] AI voice module + speaker`
- Wiring diagram




# 3. Software Preparation

**Code entry points used in this tutorial**

- Camera capture command: `/home/jetson/hardware_hal_demo/seewhat`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`

This project does not rely on a single "large project script"; instead, it has OpenClaw orchestrate multiple basic tools:

- `dht` is responsible for reading ambient temperature and humidity
- `oled` is responsible for displaying care information
- `rgb` is responsible for status alerts
- `seewhat` is responsible for observing plant appearance

# 4. Testing Basic Project Capabilities

Before starting this project, please complete an `openclaw tui` basic dialogue self-check first; if you haven't done so, see [06-Three Dialogue Configuration Tests] first. After passing, use TUI to verify the basic capabilities in the project.

Enter in terminal:

```
openclaw tui
```




After confirming OpenClaw can communicate normally, you can try the following types of commands:

- Read current temperature and humidity
- Display "Plant OK" on the OLED
- Change the RGB light to green
- Take a photo for me

If all these individual capabilities work, it means the main components required for the project are all ready.

# 5. Three Solutions for Communicating with OpenClaw

| Solution | Suitable Scenarios | Advantages | Recommended Positioning |
|---|---|---|---|
| WhatsApp | Remote care, classroom demonstrations | Easy to set up; suitable for remote status monitoring | Demonstrate "remote natural language care" |
| TUI | Local debugging, project integration | Most direct, easiest to troubleshoot | First run through multi-peripheral links |
| Voice code | Voice companionship, complete AI experience | Closer to a real assistant experience | Use as final demonstration solution |

**If the optional AI voice module + speaker is selected, Feishu dialogue, TUI, and WhatsApp dialogue need to play back OpenClaw's replies via TTS. You can run the following program in the terminal. The AI voice module dialogue itself does not need to run it again.**

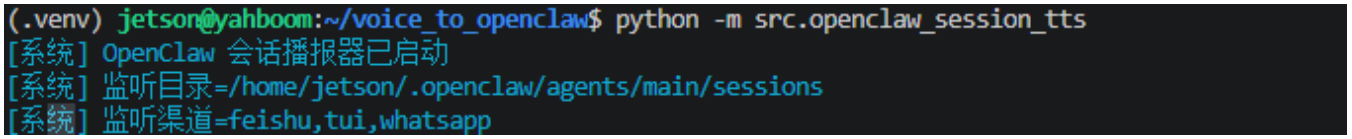
- Modify the TTS broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true  # This parameter controls whether the replies are broadcast. After modification, save it and run the following program in the terminal
```

- Enter the following in the terminal

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```




## 5.1 whatsapp  Solution

For WhatsApp integration and bot creation steps, please refer directly to [06-Three Dialogue Configuration Tests]. This section only retains dialogue examples suitable for the plant care project.

After successfully creating the bot, you can remotely inquire about plant status in WhatsApp, for example:

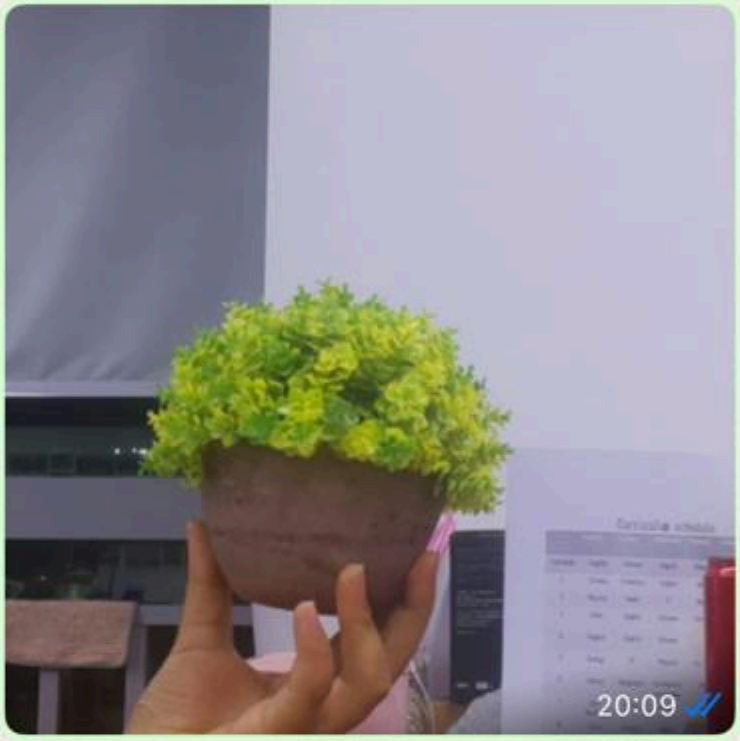
- Take a photo to check the current plant status and perform plant care

20:10

< 2 +86 153 3885 7526 (你)
给自己发消息

Take a photo to see the current plant status and perform plant care

20:09




I can see the plant! It's a small potted plant with vibrant green-yellow leaves. The foliage looks full and healthy. Let me complete the plant care workflow.

20:10

Plant care check completed!

**Status:**

- 📸 **Visual check:** Plant looks healthy with vibrant green-yellow foliage
- 🌡️ **Temperature:** 29.7°C (85.5°F)
- 💧 **Humidity:** 52%
- 📟 **OLED:** Displayed "Plant:



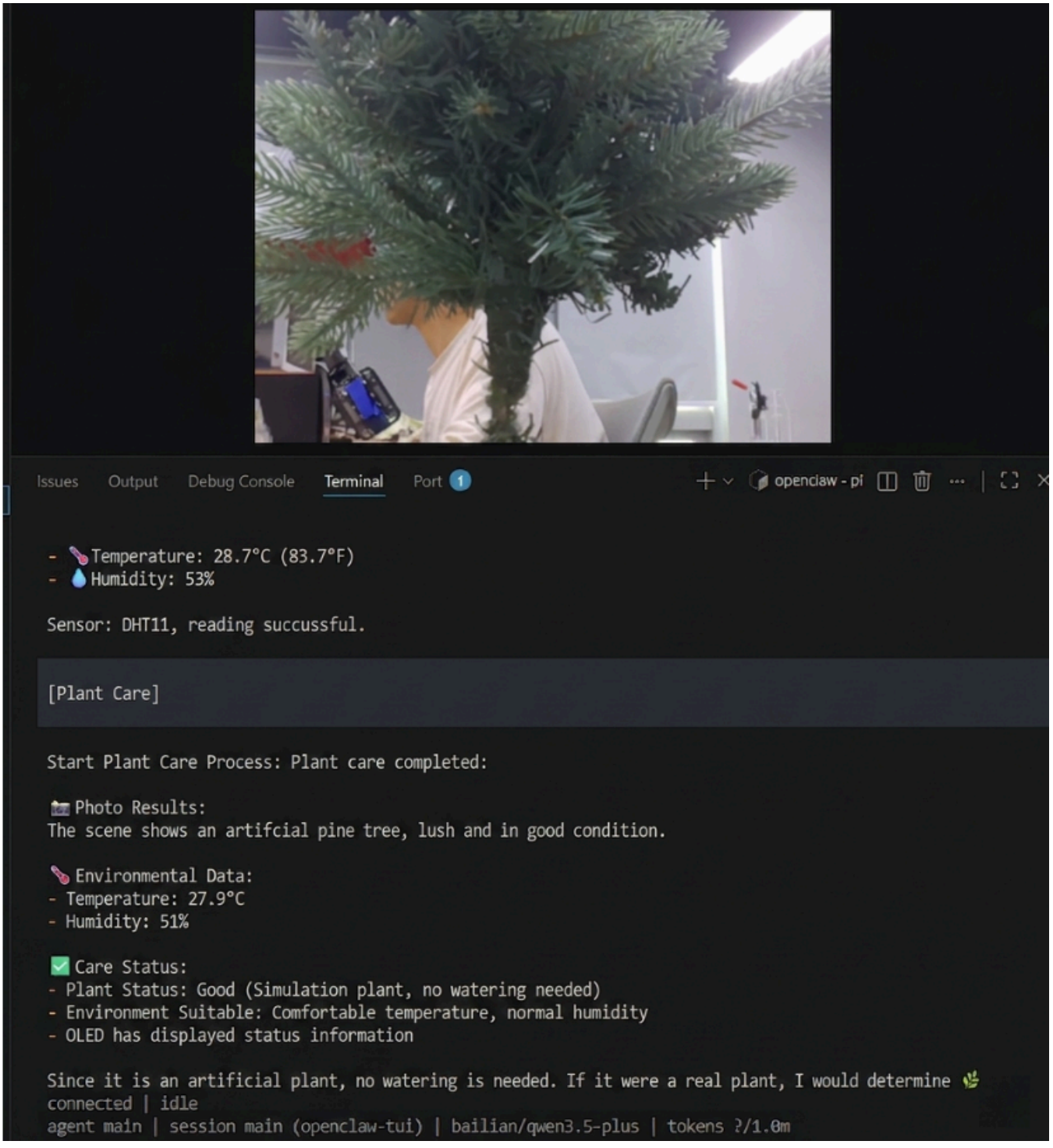
## 5.2 TUI Solution

If you have completed the TUI self-check in the Prerequisites section, directly enter terminal text dialogue:

```
openclaw tui
```

Then you can directly input project-related commands, for example:

- Plant care

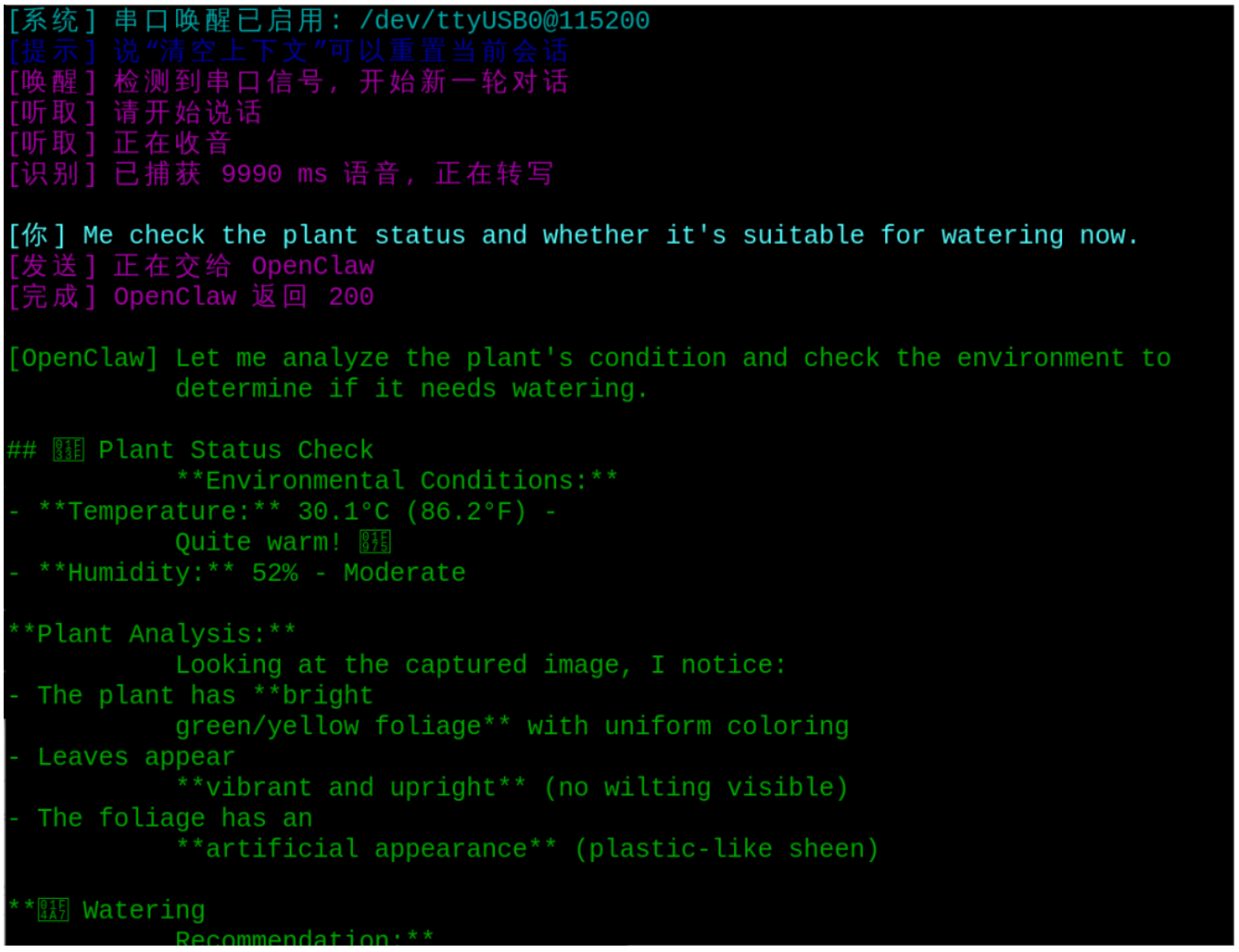


## 5.3 Voice Code Solution

```
Requires optional AI voice interaction module + speaker
```

For voice module installation, environment configuration, and basic wake-up process, please refer directly to [05-AI Voice Interaction Configuration]. After wake-up, you can directly speak control commands, for example:

- Check the plant status for me, is now a good time to water




# 6. Project Implementation

Note: The following demonstrations are all conducted using the `openclaw tui` method. You can also choose WhatsApp or voice methods to complete them.

## 6.1 Scenario 1: Environment Reading

### Experiment Objective

Read the current temperature and humidity around the plant to enable basic environmental monitoring.

### Effect Description

OpenClaw will call DHT to read current environmental data and use natural language to report the temperature and humidity.

### Example Commands

- Read the current temperature and humidity near the plant
- What are the current ambient temperature and humidity?
- Help me check if the potted plant area is too dry

## 6.2 Scenario 2: Status Judgment

### Experiment Objective

Have OpenClaw provide a brief plant care suggestion after obtaining environmental data.

### Effect Description

In addition to directly reporting values, OpenClaw can also summarize conclusions like "too dry", "too hot", or "normal status" in a more everyday language style.

### Example Commands

- Judge the plant status based on current temperature and humidity
- Is now a good time to water

## 6.3 Scenario 3: Audio-Visual Alerts

### Experiment Objective

When environmental conditions are not ideal, use OLED and RGB to provide more intuitive alerts.

### Effect Description

For example, light green when the environment is normal, yellow when too hot, red when abnormal; simultaneously display simple prompt messages on the OLED, so it's not just "talking" but also has clear peripheral feedback.

### Example Commands

- Light green if the environment is normal
- Light yellow and alert me if it's too hot
- Display "Plant Needs Care" on the OLED

### Extensible Play Ideas

- Integrate with a buzzer for stronger alerts
- Make it into a desktop care status light
- Add timed polling and auto-refresh

## 6.4 Scenario 4: Photo Observation

### Experiment Objective

Use the camera to capture the current plant image and have OpenClaw provide a simple description.

### Effect Description

This step is suitable for observing leaves, pot surfaces, or overall growth. The current tutorial focuses on "take a look and describe", not automatic disease detection, making it more suitable for teaching demonstrations and basic visual monitoring.

### Example Commands

- Take a photo of the plant
- Check the leaf condition
- Describe the current plant image after taking a photo